

Nathaniel Kessel

Loveland, Ohio

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EDUCATION

University of Cincinnati, Cincinnati, Ohio

Expected Graduation: May 2028

Bachelor of Science, Computer Engineering

Miami Valley Career Technology Center (MVCTC), Clayton, Ohio

August 2021 – June 2023

- Built foundation for troubleshooting and maintenance practices using multimeters, calipers, safety wire tooling, soldering, and other equipment

PROJECTS

ESP32 Desk Clock

September 2025 – Ongoing

Personal Project, [GitHub Repository](#)

- Engineered and iteratively debugged four hardware revisions of a custom printed circuit board (PCB) using KiCad, utilizing digital multimeters and oscilloscopes to verify signal integrity on SPI and I2C lines, power delivery, and trace routing
- Hand-soldered surface-mount (SMD) components, successfully integrating a custom SPI-driven LED display, onboard real-time clock (RTC), and NTP network time synchronization
- Prototyped circuit design and firmware in Arduino C++, with plans to add weather, calendar, and local data integration

Messaging App, *Team Leader*

May 2026 – August 2026

University of Cincinnati, [Project Website](#)

- Architected a messaging web application utilizing Scrum and agile development practices
- Designed the database schema and real-time communication pipeline slated to use Node.js, Socket.io, and MongoDB Atlas
- Established the planned deployment infrastructure via Azure App Services and GitHub Actions

Custom Headless 1U Rackmount Server & Power Supply

February 2026 – May 2026

Personal Project

- Identified power delivery constraints for 3.5" enterprise HDDs and designed a multi-stage buck converter power supply in LTspice, stepping the 19.5V input to 12V, 5V, and 3.3V rails
- Calculated inductor values and simulated transient response and voltage ripple across three cascaded buck converters operating at 300kHz switching
- Engineered a custom 1U rack-mountable chassis using Fusion360 and FDM 3D Printing to repurpose a laptop motherboard for headless system administration

SKILLS

Embedded Systems: C, C++, Arduino C++, Verilog, STM32CubeMX, ESP-IDF

Hardware & Manufacturing: KiCad 10.0, Fusion360, AutoCAD, LTspice, Multisim, SMD Soldering, 3D Printing

Software & Web: Python, HTML/CSS, JavaScript, MATLAB, VBA, LabVIEW

Cloud and Tooling: Azure App Services, Git, GitHub Actions

EXPERIENCE

Closing Expert, **Target**, Milford, Ohio

September 2023 – Present

- Balanced an average of 26 hours a week with full-time academic courseload
- Trained and guided employees on key WMS store metrics and utilizing them for improved end-of-day organization and performance metrics
- Improved fill percentage from an average of 70% to consistently over 85% during the 2024 season through updated team training and workflow optimization

AVAILABLE TO CO-OP SPRING 2027 (JANUARY – MAY)